

secondary_measure.py — Full Pipeline

Segment-Anything SAM2 → particle / fragment classification
equivalent circle diameter · sphericity (μm)

INPUT — SEM image (.jpg / .png / .tiff / ...)
single file or batch directory

IMAGE NORMALISATION
resize → 2048 × 1636 (bilinear) · crop bottom 100 px → 2048 × 1536

ROI EXTRACTION
--roi_x_min/y_min/x_max/y_max (default: full normalised image)
or auto_center_crop if enabled

TILE GRID
ROI → overlapping tiles (tile_size=512 × stride=256)
edge tiles extended · duplicates deduplicated

TEXTURE ENHANCEMENT (per tile)
CLAHE + Sobel gradient + Laplacian + morphological blackhat
→ single enhanced grayscale channel

INTEREST-POINT SAMPLING (Shi-Tomasi)
cv2.goodFeaturesToTrack on enhanced tile
fallback: Otsu → contour centroids (if sparse)
up to points_per_tile=80, spacing ≥ point_min_distance=14 px

SAM2 BATCH INFERENCE (Ultralytics)
foreground point prompts (label = 1), batch_size = 32 points per call
raw logit masks → binarise at mask_binarize_threshold = 0.0

BOUNDING-BOX IoU DEDUP (tile-level, fast)
reject mask if bbox IoU ≥ 0.85 with any accepted mask
promote tile-coord mask → ROI-coord mask

PIXEL IoU DEDUP (ROI-level)
binary mask IoU ≥ 0.60 → reject as duplicate

MASK REFINEMENT (morphology, optional)
morphological open (remove small spurs)
morphological close (fill interior holes)

MEASUREMENT
eq_diameter_um = $2 \times \sqrt{(\text{mask_area} / \pi)}$ → px → μm (Equivalent Circle Diameter)
sphericity = $4\pi \times \text{mask_area} / \text{perimeter}^2$ (Wadell 2D isoperimetric ratio)
--eq_diameter ON (default): size = eq_diameter | OFF: size = (spanH + spanV) / 2

CLASSIFICATION
mask_area < area_threshold (1500 px²) → fragment
otherwise → particle

OUTPUTS (per image)
01...04 *.png overlay | objects.csv particles.csv fragment_masks/ particle_masks/
particle_dist.png sphericity_dist.png | summary.json objects.json debug.json
Batch: img_id_summary.json · batch_summary.json